

Sleep-HRV Interaction Analysis Using Wearable Data

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Introduction

Heart rate variability (HRV) reflects variation between consecutive heartbeats and is often used as an indicator of recovery, stress and autonomic nervous system regulation. Sleep may influence HRV because deep sleep, REM sleep and sleep efficiency are related to physiological recovery. This project analyses wearable sleep and vital-sign data to determine which sleep metrics are most informative for next-day HRV.

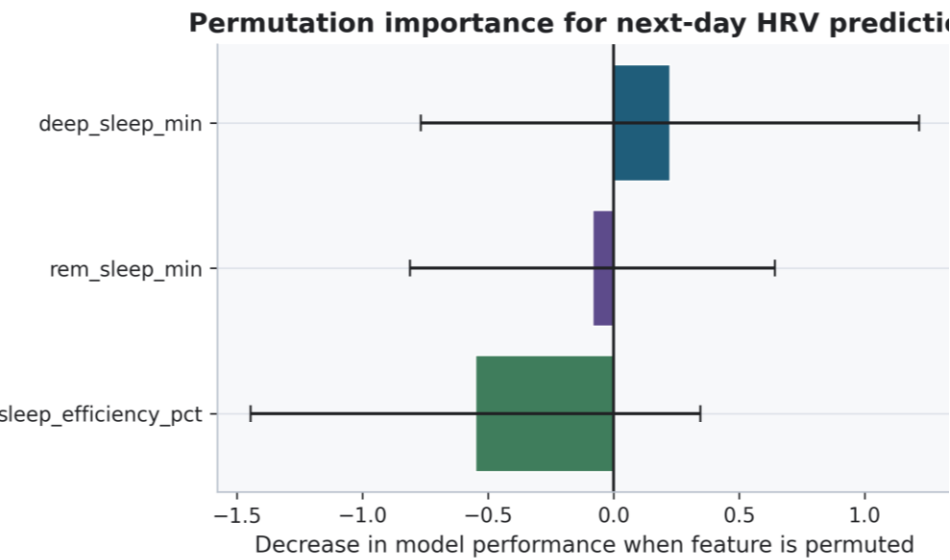


Figure 1. Permutation importance showing which required sleep metric contributed most to next-day HRV prediction.

Method

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- Wearable Sleep, Vital Signs and Activity CSV files were cleaned and merged by date.
- Sleep records were aggregated by wake-up date.
- Previous-day activity was aligned with next-day HRV.
- Missing values, duplicates and outliers were checked using IQR and Z-score methods.
- The main task was regression because next-day HRV is continuous.
- Models were evaluated using a temporal train-validation-test split and 5-fold time-series cross-validation.

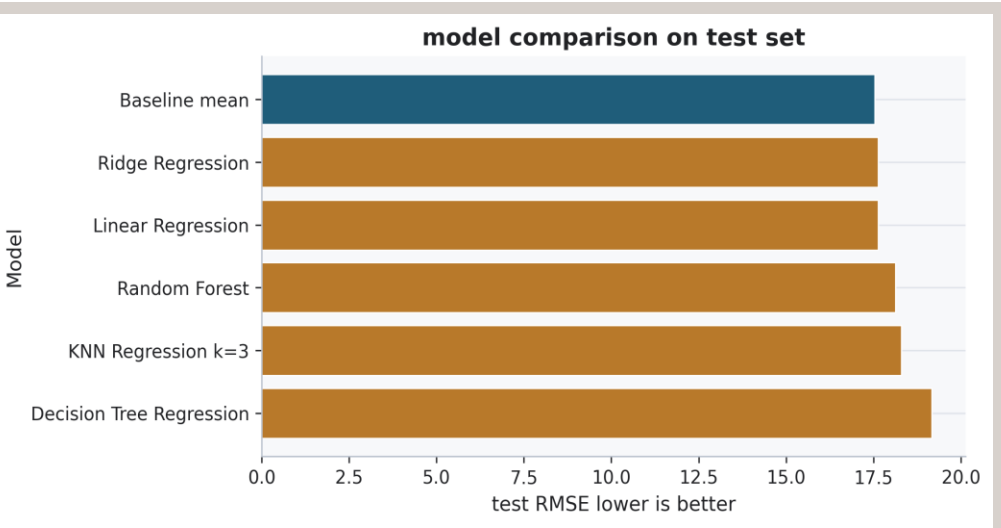


Figure 2. Test RMSE comparison of regression models. Lower RMSE indicates better prediction performance.

Results

Final selected model:
Distance-weighted KNN regression, k = 3

Final temporal test set:
MAE = 18.578 ms
RMSE = 19.447 ms
 $R^2 = -62.734$ The negative R^2 shows that the model did not generalise well to unseen future data.

Time-series cross-validation:
Ridge Regression: RMSE = 13.655
Linear Regression: RMSE = 13.662
KNN Regression k = 5: RMSE = 13.929
Decision Tree: RMSE = 14.863

The full feature ranking showed that average heart rate had the strongest association with next-day HRV:
 $r = -0.923$

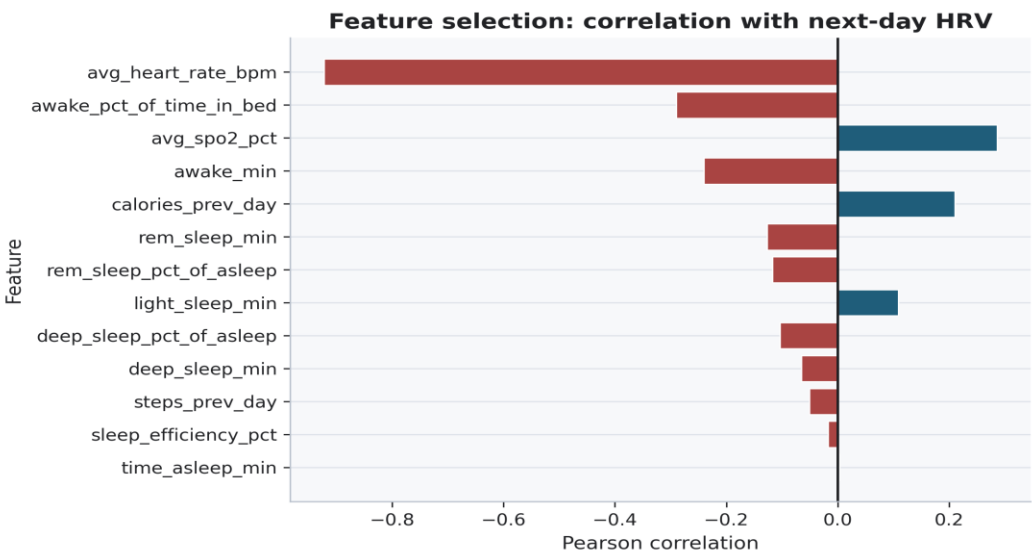


Figure 4. Feature correlation ranking showing associations between candidate variables and next-day HRV.

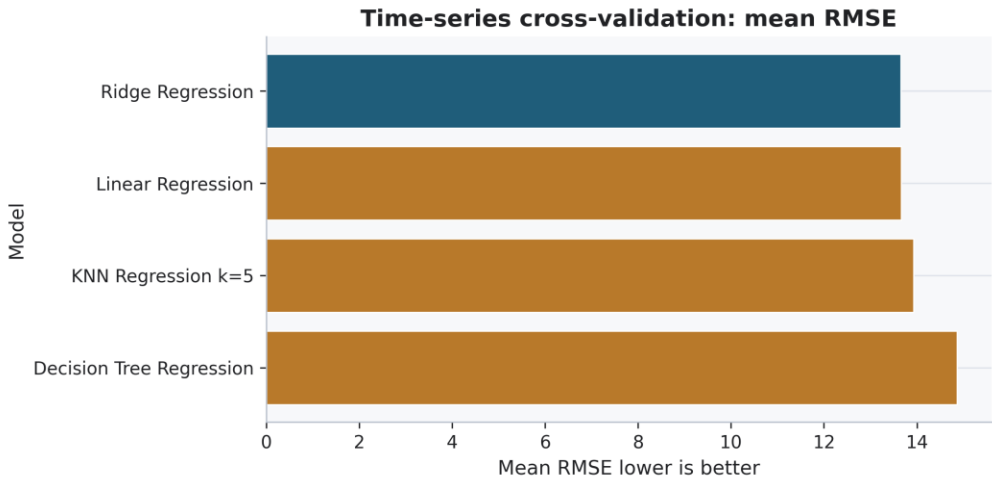


Figure 3. Mean RMSE from 5-fold time-series cross-validation used as an additional robustness check.

Interpretations

The sleep-only model was not strong enough for reliable HRV prediction. This suggests that deep sleep, REM sleep and sleep efficiency alone do not fully explain next-day HRV in this dataset.

Average heart rate was much more strongly related to HRV than the sleep variables which is expected. However, the main model stayed focused on sleep metrics because that was the project task.

Among the required sleep metrics, deep sleep duration was the most informative feature in the combined influence analysis.

The results should be read as exploratory wearable-data findings, not as clinical or causal conclusions.

Conclusions

This project developed a complete wearable-data pipeline for analysing the relationship between sleep metrics and next-day HRV.

The final KNN regression model with k = 3 was selected during validation, but the negative test R^2 showed poor generalisation to future data.

Therefore, the model should not be used as a clinical prediction tool. Its value is mainly exploratory.

The main conclusion is that deep sleep duration was the most informative required sleep metric, while HRV also appeared strongly connected to broader physiological variables such as average heart rate.

Data and tools

Data: Wearable Sleep, Vital Signs and Activity CSV files

Tools: Python, pandas, NumPy, scikit-learn and matplotlib

Main methods: Temporal validation, time-series cross-validation, KNN regression, correlation analysis and permutation importance